Data Structures

BS (CS) _Fall_2025

Lab_02 Task

Learning Objectives:

1. Class Templates with unit testing (Gtest)

Class Templates in C++

Task 0: Simple Generic Holder

Create a template class Holder<T> that stores a single item of any type.

Data Members:

- **T item :** the stored value.

Functions:

- **Constructor:** initialize the item.
- **setItem(T value):** set the stored item.
- **getItem():** return the stored item.
- **displayType():** prints the data type of the stored item using typeid(item).name() from typeinfo header.

Test Cases:

- Create Holder<int>, Holder<double>, and Holder<string>.
- Set and get values for each.
- Print stored type and value.

Task 1:

You have to make a class for an online Banking system. The bank keeps a record of all the transactions made along with the current balance. The bank allows a specific number of transactions to be made.

You have to define a template class **OnlineBank** which can store different types of balance e.g. int, double, float etc. Bank will have same type of elements at a time. The **OnlineBank** class should have the following member variables:

- **T currentBalance** the current balance in the bank.
- **T* withdrawals** A pointers to set the withdrawals made from the account. Before making a withdrawal, it should be checked if it's smaller than or equal to the current balance.
- **T* deposits** A pointers to set the deposits made from the account.
- **numOfTransactions** this is the number of withdrawals and deposits that can be made
- **counterWithdrawal** this is the number of current withdrawals made

- **counterDeposits** this is the number of current deposits made

The class should consist of following functions:

- **Constructor** with current balance and number of transactions as parameter
- **isWithdrawal** checks if the number of current withdrawals is less than the numOfTransactions
- **isDeposit** checks if the number of current deposits is less than the numOfTransactions
- **makeWithdrawal** which will return true if the value passed in the parameter is less than or equal to the current balance and its counter is less than the limit. It will deduct the given amount from current balance and update it.
- **makeDeposit** which will get a value to be added in the current balance if its counter is less than the limit
- **getCurrentBalance()** will return the current balance
- **print** this will print the current balance

Test Cases:

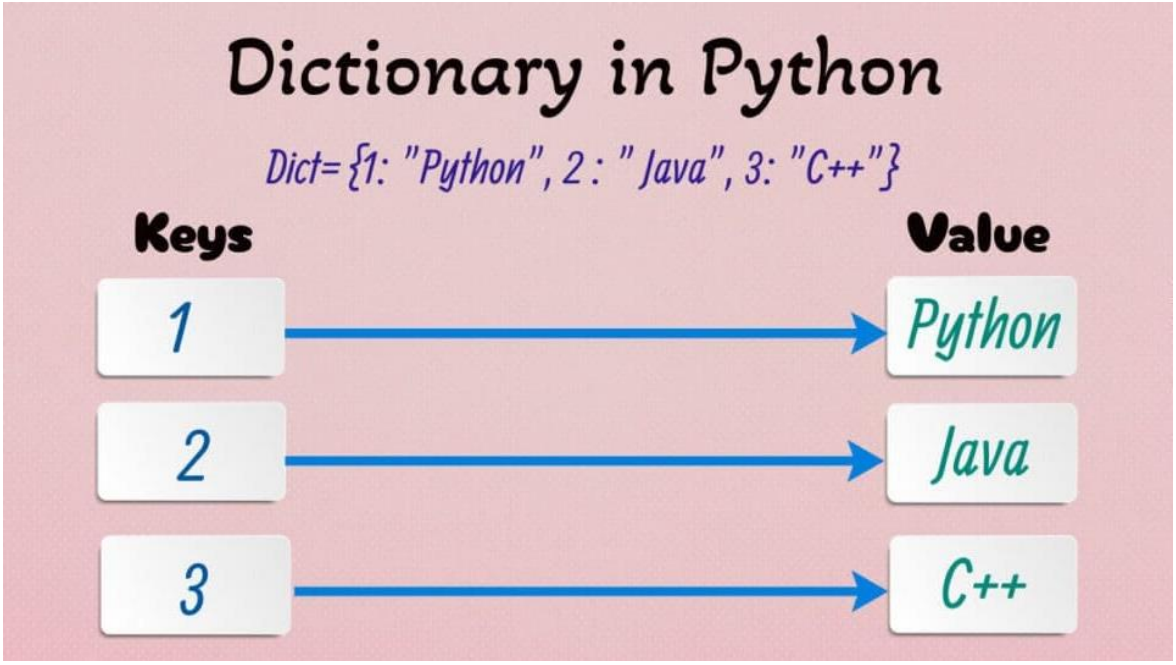
Create a comprehensive test class for the **OnlineBank** template class. Create separate instances for **int, float, double** data types and perform the following test cases:

- Create OnlineBank instances with different initial balances and transaction limits
- Test multiple valid deposits within transaction limits and verify balance updates
- Test multiple valid withdrawals where amount ≤ currentBalance and verify balance updates
- Attempt invalid withdrawals where amount > currentBalance and verify rejection
- Test transaction limits by exceeding both deposit and withdrawal limits
- Perform mixed deposits and withdrawals in random combinations
- Test boundary cases like zero balance, minimum amounts, and exact balance withdrawals
- Test data type specific operations (whole numbers for int, decimals for float/double)

Task 2:

Template Class Dictionary:

You are designing a generic Dictionary class similar to Python's dict that stores key–value pairs of any type.



The dictionary should be flexible enough to:

- Work with any combination of key and value types.
- Allow capacity to be set at runtime when the object is created.
- Provide default types for convenience.
- Have a special behavior when keys are std::string → make it case-insensitive by automatically converting keys to lowercase and performing searches/removals ignoring case.

The class will have Three template parameters:

- K for the key type (default: int)
- V for the value type (default: string)

Data Members (Private)

- **K* keys;** :- dynamic array storing keys.
- **V* values;** :- dynamic array storing corresponding values.
- **int count;** :- current number of elements.
- **int currentSize;** :- current allocated size of arrays (starts small, doubles when needed).

Member Functions (Public)

Constructor

- Initializes count = 0, currentSize = 2.
- Allocates keys = new K[currentSize] and values = new V[currentSize].

Destructor

- Deletes dynamically allocated arrays.

isEmpty()

- Returns true if count == 0.

insert(K key, V value)

- If the key already exists, do nothing.
- If count == currentSize, reallocate arrays with double the size.
- Insert the new key–value pair.

search(K key)

- Returns true if key exists, otherwise false.

remove(K key)

- If found, remove the key–value pair and shift the rest of the elements.

getValue(K key)

- Returns the value if key exists, otherwise throws an exception.

print()

- Prints all key–value pairs.

Test cases :

1. **General Dictionary Tests (Default Parameters)(K = int, V = std::string)**
   - Create Dictionary<>.
   - Insert 3+ key value pairs:
     - Example: 1 : "Ali", 2 :"Sara", 3 : "John".Insert duplicate key (e.g., 2 : "Sara") :- should not add.
   - Search existing key (2) :- returns true.
   - Search non-existing key (10) :- returns false.

- Get value of existing key (2) :- returns "Sara".
- Remove key (2) :- confirm it’s gone.
- Print dictionary when empty and when not empty.

2. **Auto resizing Tests : (K = int, V = double)**
   - Create Dictionary<int, double>.
   - Insert more elements than initial size (e.g., > 2) with values like 101 : 90.5, 102 : 85.3, etc.
   - Verify resizing happens and all inserted elements remain accessible.
3. **String Key Specialization Tests (Case Insensitive)(K = std::string, V = int)**
   - Create Dictionary<std::string, int>.
   - Insert "Apple" : 5, "Banana" : 10.
   - Search "APPLE", "apple", "ApPlE" : all return true.
   - Remove "APPLE" , confirm "apple" is removed.
4. **Different Data Type Combination Tests**
   i) **Double keys & char values**
      - Create Dictionary<double, char>.
      - Insert 1.1 : 'A', 2.2 : 'B'.
      - Search 2.2 , returns true.
      - Remove 1.1 , confirm removal.

   ii) **Char keys & bool values**
      - Create Dictionary<char, bool>.
      - Insert 'x' : true, 'y' : false.
      - Search 'y' , returns true.
      - Get value 'x' , returns true.